

The five theorems of modular Koszul duality

Raez Lorgat

Abstract

Koszul duality over a point closes through a quadratic coalgebra, the full bar construction, and cobar reconstruction. Over an algebraic curve, Ran collisions and the geometry of moduli add further layers. The enhanced bar differential in $D(\text{Ran}X)$ is square-zero. After second-kind cobar realization, its fiberwise modular algebra is curved, $m_1^2 = [m_0, -]$. A normalized collision trace extracts the scalar $\kappa(\mathcal{A})$, while the full curvature retains operator-valued data.

The chiral algebra $\mathcal{A}$ is the level-1 chart $A_b = \text{End}_C(b)$ of a factorization dg-category C^{op} on a tangential log curve (X, D, τ) , evaluated at a vacuum object b . Five theorems characterize the levels of the resulting Beilinson diagram

$$\begin{array}{ccccc} C^{\text{op}} & \xrightarrow{b} & A_b & \xrightarrow{\bar{B}} & B_X(A_b) \\ & & \downarrow Z_{\text{ch}}^{\text{der}} & & \downarrow \mathbb{D}_{\text{Ran}} \\ & & C_{\text{ch}}^{\bullet}(A_b, A_b) & & K_X(A_b). \end{array}$$

Here $C_{\text{ch}}^{\bullet}(A_b, A_b) := R\text{Hom}_{A_b \otimes A_b^{\text{op}}}(A_b, A_b)$ is the algebraic derived centre in the completed chiral bimodule category. A physical bulk realization consists of an open–closed equivalence

$$\Psi_{\text{OCA}} : \mathcal{O}_{\text{bulk}}^{\text{phys}}(A_b) \xrightarrow{\sim} C_{\text{ch}}^{\bullet}(A_b, A_b).$$

The line-operator and modular-trace stages carry their respective comparison data. Each theorem carries a type signature (*quadrant, presentation, level, hypothesis package*).

Theorem A separates Ran reconstruction $R_X(\mathcal{A}) = \Omega_X B_X(\mathcal{A}) \xrightarrow{\sim} \mathcal{A}$ from the Verdier–Koszul algebra $K_X(\mathcal{A}) = \mathbb{D}_{\text{Ran}} B_X(\mathcal{A})$. The ambient Ran equivalence is the theorem of Francis–Gaitsgory; its restriction to the chosen factorization subcategories carries the package H_{fact} . Theorem B is the quadratic recognition theorem: under H_{CL} , the map $q_{\mathcal{A}} : \mathcal{A}^1 \rightarrow B_X(\mathcal{A})$ is a quasi-isomorphism precisely when the associated twisted tensor products are acyclic and the quadratic bar spectral sequence converges with diagonal support. Theorem C is the 5×5 κ -stratification matrix on rows G/L/C/M/B with columns $\{\kappa_{\text{cat}}, \kappa_{\text{ch}}^{\text{Hodge}}, \kappa_{\text{ch}}^{\text{Heis}}, \kappa_{\text{BKM}}, \kappa_{\text{fiber}}\}$ at level 5 per stratum; $K_3 \times E$ has native entries (o, o, ★, 5, 24), and its Open Heisenberg entry 3 is conditional on the realization map $\eta_{K_3, E}$. On a family carrying the represented anti-invariant pairing in H_C , the Lagrangian decomposition yields the corresponding duality-orbit sum; the standard Kac–Moody, free-field, and Virasoro traces give o, o, and 13 on their respective comparison surfaces. Theorem D separates four typed objects: $\text{Obs}_g^{\text{def}} \in H^2(\text{Def}_g)$ for $g \geq 2$, $\mathfrak{D}_g^K = \kappa \lambda_{-1}(\mathbb{E}_g)$ under H_D^K for $g \geq 2$, $\text{obs}_g^{\text{Hdg}} = (-1)^g \kappa \lambda_g$ for $g \geq 2$, and $F_g = \kappa \lambda_g^{\text{FP}} + \partial F_g^{\text{cross}}$ under the stable-graph package. The genus-one trace has the normalization $\text{tr}_1 \text{Obs}_{1,1}^{\text{def}} = \kappa \lambda_1$. Theorem H is family-indexed: a strong deformation retract $H_H(A_b; S)$ identifies the chiral Hochschild complex with a model supported in S . The bounded Virasoro and rank-one even superboson calculations enter the chart complex through a specified comparison quasi-isomorphism.

The ordered bar complex $\bar{B}_X^{\text{ord}}(A_b)$ with its deconcatenation coproduct and collision kernel $K_{A_b}^{\text{coll}}(z)$ is a common source for the shadow operators. Their scalar coordinates use normalized

equivariant functionals. In arity two, $q_2(K_{\mathcal{A}}^{\text{coll}}) \in \text{End}_{\mathcal{A}}(2)_{h\Sigma_2}$, and $\tau_{\mathcal{A},2}^{\text{res}} : \text{End}_{\mathcal{A}}(2)_{h\Sigma_2} \rightarrow \mathbb{C}$ recovers κ on the abelian and scalar lanes; the non-abelian Kac–Moody value $\kappa = \tau_{\mathcal{A},2}^{\text{res}} q_2(K_{\mathcal{A}}^{\text{coll}}) + \dim(\mathfrak{g})/2$ encodes the Sugawara shift. The locus $\text{KSDual} \subset \text{Kosz}(X)$ on which the Verdier involution $\mathcal{A} \mapsto \mathcal{A}^!$ acts by the identity is the $\mathbb{Z}/2$ -fixed sublocus in the finite-type curved Verdier–Koszul ambient. On its admissible Koszul-complete part, a chosen self-duality map $\sigma_{\mathcal{A}} : \mathcal{A} \xrightarrow{\sim} K_X(\mathcal{A})$ represents the fixed point, and Theorem H retains its family support set S . The Borchers trace identity [5, 14] $\kappa_{\text{BKM}}(\Phi_N) = c_N(o)/2$ is the equivariant signature. The Heisenberg algebra provides the first exact local calculation and displays the comparison maps required for its global realization.

Contents

1	Introduction: from a point to a curve	3
1.1	The problem	3
1.2	The answer	3
1.3	Prior work	5
1.4	The Heisenberg computation	5
1.5	Conventions	6
2	The bar complex on algebraic curves	6
2.1	Chiral algebras	6
2.2	The propagator and the Arnold relation	7
2.3	The ordered bar complex	7
2.4	Why $d^2 = 0$: the Borchers selection principle	8
2.5	Two coproducts	8
2.6	From genus 0 to all genera	8
3	Theorem A: Ran reconstruction and Verdier refinement	9
3.1	Statement	9
3.2	Proof	10
4	Theorem B: quadratic Koszul recognition	11
4.1	Statement	11
4.2	Proof architecture	11
5	Theorem C: the 5×5 stratification atlas	12
5.1	Five measurements	12
5.2	Three comparison tiers	12
5.3	Statement	13
6	Theorem D: the modular characteristic hierarchy	14
6.1	The four typed objects	14
6.2	Statement	15
6.3	The Chern-root calculation	16
6.4	The Faber–Pandharipande scalar	16

7	Theorem H: chiral Hochschild cohomology	17
7.1	Statement	17
7.2	Family-specific specializations	18
7.3	Proof architecture	18
8	The dependency graph among Theorems A, B, C, D, and H	18
8.1	One new input per theorem	20
9	The Heisenberg chart: the first complete local calculation	20
9.1	Exact local algebra	20
9.2	Theorem A: full Ran reconstruction	20
9.3	Theorem B: the finite-window comparison	21
9.4	Theorems C and D: scalar and Hodge realizations	21
9.5	Theorem H: bounded benchmark and chart transport	22
9.6	Summary of the worked case	22
10	The standard-family atlas: source data and comparison status	22
10.1	Local calculations before global transport	22
10.2	The census-scoped archetypes	23
10.3	The Virasoro scalar lane	24
10.4	Family-indexed Hochschild support	24
10.5	Provenance rule for every row	25
A	Comparison morphisms for the five theorems	26
B	Exact computational layer	27
C	Open construction problems	27

I Introduction: from a point to a curve

I.1 The problem

Koszul duality over a point begins with the full coalgebra $B(A) = (T^c(s^{-1}\tilde{A}), d_B)$ and the reconstruction counit $\Omega B(A) \rightarrow A$ [3, 15, 21]. A connected quadratic presentation supplies $A^i \rightarrow B(A)$; the Koszul criterion identifies this map as a quasi-isomorphism and forms $A^! = (A^i)^\vee$.

Over an algebraic curve, the generators are sections of a $\mathcal{D}_X$ -module, and the relations are operator product expansions with arbitrary pole order. Higher-genus geometry places the fiberwise cobar algebra in a curved A_∞ /CDG package: the raw identity is $m_1^2 = [m_0^{(g)}, -]$, and its scalar diagonal projection is $\kappa(\mathcal{A}) \cdot \omega_g$.

The question is: what structure replaces the terminating classical theory?

I.2 The answer

The chiral algebra $\mathcal{A} = \mathcal{A}_b$ is the chart at a vacuum $b \in C^{\text{op}}$ of a factorization dg-category on a tangential log curve (X, D, τ) . Its two algebraic branches are the Ran bar–Verdier diagram $\mathcal{A}_b \xrightarrow{\bar{B}} B_X(\mathcal{A}_b) \xrightarrow{\mathbb{D}_{\text{Ran}}}$

$K_X(A_b)$ and the derived-centre construction $A_b \mapsto C_{\text{ch}}^\bullet(A_b, A_b)$. These branches organise levels 1–3 of the open quadrant. The open–closed equivalence Ψ_{OCA} supplies a physical bulk realization. On the strictly augmented lane, the ordered bar complex

$$\overline{B}_X^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}}), \quad |s^{-1}v| = |v| - 1,$$

with deconcatenation coproduct

$$\Delta[s^{-1}a_1 | \cdots | s^{-1}a_n] = \sum_{i=0}^n [s^{-1}a_1 | \cdots | s^{-1}a_i] \otimes [s^{-1}a_{i+1} | \cdots | s^{-1}a_n], \quad (\text{I})$$

is a cofree E_1 -coalgebra. Here $\tilde{\mathcal{A}} = \ker(\varepsilon: \mathcal{A} \rightarrow \omega_X)$ is the augmentation ideal and s^{-1} is the desuspension, lowering degree by 1. The integral kernel is the logarithmic 1-form

$$\eta_{12} = d \log(z_1 - z_2) = \frac{dz_1 - dz_2}{z_1 - z_2},$$

has residue 1. A coordinate change adds an exact logarithmic form regular along the diagonal, so the residue class is intrinsic. At three points, Arnold’s relation [1]

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$$

generates all relations in $H^*(\text{Conf}_n(\mathbf{C}), \mathbf{C})$. Under the faithful Fulton–MacPherson jet realization, bar nilpotence and the Borchers identity are equivalent: Arnold’s relation governs the boundary forms, and the Borchers identity governs their OPE coefficients. This is the *selection principle*: configuration-space geometry realizes the chiral identities at every collision stratum.

The binary collision residue of the bar differential is the collision kernel

$$r_{\mathcal{A}}^{\text{coll}}(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}}) \in \text{End}_{\mathcal{A}}(2)[[z^{-1}]],$$

where $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is the Maurer–Cartan element of the bar differential, satisfying $d\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ automatically because $(D_{\mathcal{A}})^2 = 0$. A classical r -matrix interpretation requires a Yang–Baxter identity in a specified representation. Homotopy coinvariants retain the equivariant endomorphism data. A normalized trace functional

$$\tau_{\mathcal{A}}: \text{End}_{\mathcal{A}}(2)_{h\Sigma_2} \longrightarrow \mathbf{C}$$

defines the scalar

$$\kappa(\mathcal{A}) = \tau_{\mathcal{A}}(r_{\mathcal{A}}^{\text{coll}}) \quad \text{for abelian and scalar families,}$$

while for non-abelian affine Kac–Moody

$$\kappa(V_k(\mathfrak{g})) = \tau_{V_k(\mathfrak{g})}(r_{V_k(\mathfrak{g})}^{\text{coll}}) + \dim(\mathfrak{g})/2.$$

This is the modular characteristic selected by the trace. Five structural theorems extract the Σ_n -invariant content of the ordered bar. Each theorem characterizes one facet of $\Theta_{\mathcal{A}}$:

- (A) *Reconstruction and Verdier refinement*: $R_X(\mathcal{A}) = \Omega_X B_X(\mathcal{A}) \xrightarrow{\sim} \mathcal{A}$ and $K_X(\mathcal{A}) = \mathbb{D}_{\text{Ran}} B_X(\mathcal{A})$ at levels $1 \leftrightarrow 2$.

- (B) *Quadratic recognition*: $q_{\mathcal{A}}: \mathcal{A}^i \xrightarrow{\sim} B_X(\mathcal{A})$ under H_{CL} .
- (C) *Stratification*: the 5×5 κ -stratification matrix on rows G/L/C/M/B at level 5 per stratum. The $K_3 \times E$ row has native entries (o, o, $\star$, 5, 24), and H_{B} supplies its Open Heisenberg entry 3.
- (D) *Modular hierarchy*: deformation class, virtual K -object, signed Hodge shadow, and numerical graph trace, connected by the packages H_D^1 , H_D^K , H_D^{tr} , and H_D^{graph} .
- (E) *Family support*: $H_H(\mathcal{A}; S)$ gives $\text{Supp ChirHoch}^\bullet(\mathcal{A}) \subseteq S$ at level 3.

The chain is typed: A constructs reconstruction and the Verdier branch; B recognizes the quadratic model; C organizes the five scalar measurements; D compares deformation, Hodge, and graph realizations; H computes family-specific cohomological support.

The locus $\text{KSDual} \subset \text{Kosz}(X)$ on which the Verdier involution $\mathcal{A} \mapsto \mathcal{A}^!$ acts by the identity is the $\mathbb{Z}/2$ -fixed sublocus where the four-quadrant grid $(\text{Open} \cup \text{CY}) \times (\text{Cat} \cup \text{Chain})$ degenerates into self-dual form: Theorem A supplies the Ran unit and counit together with the separate Verdier branch; Theorem H supplies the family set S under $H_H(\mathcal{A}; S)$; the five-archetype label is rigid on the admissible finite-type curved fixed locus, and $\kappa_{\text{BKM}}(\Phi_N) = c_N(\text{o})/2$ [5, 14] is the equivariant signature for $N \in \{1, 2, 3, 4, 6\}$ plus half- and quarter-integer continuations. Scalar anti-diagonal relations such as $K = \text{o}$ or $c = 13$ are duality-orbit sums; the finite-type curved Verdier–Koszul fixed-point criterion supplies KSDual membership.

1.3 Prior work

Five existing programmes supply distinct layers of the problem. Beilinson–Drinfeld [2] construct chiral and factorization algebras on $\text{Ran}(X)$ together with chiral homology. Francis–Gaitsgory [7] establish enhanced Koszul duality in the pro-nilpotent Ran category. Costello–Gwilliam [6] develop factorization algebras for perturbative quantum field theory. Gui–Li–Zeng [13] prove a quadratic duality at the cochain level, a genus-0 degree-2 result. Malikov–Schechtman [20] construct homotopy chiral algebras as the appropriate derived input. The five theorems below organize the comparison maps among these layers.

	A	B	C	D	H
Programme	adjunction	inversion	compl.	genus	chiral Hochschild
BD [2]	D				
FG [7]	P				
CG [6]					
GLZ [13]		$P_{g=0, d=2}$			
MS [20]		D			
This work	$P + Q$	C	C	C	C

P : proved in the cited source; D : foundational or definitional input; C : conditional on the displayed comparison package; Q : factorization restriction presently conjectural. In column A, $P + Q$ records the published Ran equivalence together with its conjectural restriction to the selected factorization subcategory.

1.4 The Heisenberg computation

The Heisenberg chiral algebra $\mathcal{H}_k$ is generated by a single current J of conformal weight 1 with OPE $J(z)J(w) \sim k/(z-w)^2$. The normalized coefficient of the double pole is

$$\kappa(\mathcal{H}_k) = k.$$

At $k = 0$, Theorem A reconstructs the chart from its ordinary Ran bar. At $k \neq 0$, the second-kind package H_{curv} supplies the curved coderived counit. A quadratic Heisenberg comparison is an instance of Theorem B once the finite-window contraction in $H_{\text{CL}}^{\text{II}}$ is supplied. Under the packages of Theorem D,

$$\text{tr}_1 \text{Obs}_{\text{I},1}^{\text{def}}(\mathcal{H}_k) = k\lambda_1, \quad \text{obs}_g^{\text{Hdg}}(\mathcal{H}_k) = (-1)^g k\lambda_g, \quad g \geq 2,$$

and the scalar-diagonal graph functional has generating series $k((x/2)/\sin(x/2) - 1)$. For the rank-one even free superboson, Bakalov–De Sole–Kac compute the bounded vector $(2, 1, 0, \dots)$; a bounded-to-chart quasi-isomorphism transports this vector to Theorem H. Thus the rank-one example fixes the scalar normalization and displays every comparison map that remains family-dependent.

1.5 Conventions

Throughout, X denotes a smooth projective curve over $\mathbf{C}$, $\mathcal{A}$ a chiral algebra on X in the sense of Beilinson–Drinfeld [2]. The stack $\overline{\mathcal{M}}_{g,n}$ is used in the stable range $2g - 2 + n > 0$; we write $\overline{\mathcal{M}}_g = \overline{\mathcal{M}}_{g,0}$ for $g \geq 2$. Grading is cohomological: $|d| = +1$. Desuspension lowers degree: $|\varsigma^{-1}v| = |v| - 1$. On the strictly augmented lane, the bar complex uses $\overline{B}(\mathcal{A}) = T^c(\varsigma^{-1}\tilde{\mathcal{A}})$ with $\tilde{\mathcal{A}} = \ker(\varepsilon)$. On a centrally curved lane, a vacuum splitting replaces the algebra augmentation. The second-kind bar coalgebra carries a degree-two curvature functional h_B ; its curved cobar algebra carries the corresponding element m_o . The algebraic Fulton–MacPherson compactification $\overline{\text{FM}}_n(X)$ is the iterated algebraic blowup of X^n along the diagonal building set [8]. It is smooth, and its boundary is a simple normal-crossings divisor with components D_S indexed by subsets $S \subseteq \{1, \dots, n\}$ of cardinality at least 2. The oriented real blowup of its complex-analytic boundary is the Betti model with corners. The algebraic bar uses the first space; its configuration-space integrals may be realized on the second.

2 The bar complex on algebraic curves

2.1 Chiral algebras

A *chiral algebra* $\mathcal{A}$ on a smooth curve X [2] is a $\mathcal{D}_X$ -module equipped with a chiral bracket

$$\mu^{\text{ch}}: j_* j^*(\mathcal{A} \boxtimes \mathcal{A}) \longrightarrow \Delta_!(\mathcal{A}),$$

where $j: X^2 \setminus \Delta \hookrightarrow X^2$ is the complement of the diagonal and $\Delta_!$ is the $!$ -pushforward along the diagonal embedding. In local coordinates, the chiral bracket encodes the operator product expansion:

$$a(z)b(w) \sim \sum_{n \geq 0} \frac{a_{(n)}b(w)}{(z-w)^{n+1}}.$$

The zeroth product $a_{(0)}b$ is the Lie bracket (first-order pole); the first product $a_{(1)}b$ carries the symmetric inner product (second-order pole). The Borcherds identity

$$\sum_{j \geq 0} \binom{m}{j} (a_{(n+j)}b)_{(m+k-j)}c = \sum_{j \geq 0} (-1)^j \binom{n}{j} \left(a_{(m+n-j)}(b_{(k+j)}c) - (-1)^n b_{(n+k-j)}(a_{(m+j)}c) \right) \quad (2)$$

is the defining axiom: it couples all pole orders in a single constraint.

2.2 The propagator and the Arnold relation

The bar differential extracts OPE data from collisions on $C_2(X) = X \times X \setminus \Delta$. Poincaré residue acts on logarithmic first-order poles. The canonical local generator is the logarithmic derivative

$$\eta_{12} = d \log(z_1 - z_2) = \frac{dz_1 - dz_2}{z_1 - z_2}, \quad \text{Res}_\Delta \eta_{12} = \mathbf{1}.$$

This is a local logarithmic generator. Its residue is independent of the chosen coordinate.

Under $z \mapsto w(z)$, the local generator transforms as $\eta_{12} \mapsto \eta_{12} + d \log \frac{w(z_1) - w(z_2)}{z_1 - z_2}$; near the diagonal the second summand is regular. Thus the class of η_{12} in the logarithmic normal quotient, and hence its Poincaré residue, glues canonically; a global propagator requires the additional choice of a connection or prime-form normalization.

At three points, Arnold's relation [1]

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$$

restates the partial-fractions identity

$$\frac{\mathbf{1}}{(z_1 - z_2)(z_2 - z_3)} + \frac{\mathbf{1}}{(z_2 - z_3)(z_3 - z_1)} + \frac{\mathbf{1}}{(z_3 - z_1)(z_1 - z_2)} = 0.$$

Arnold proved that, replicated across all triples $\{i, j, k\} \subset \{1, \dots, n\}$, this generates the complete ideal of relations in $H^*(\text{Conf}_n(\mathbf{C}), \mathbf{C})$.

2.3 The ordered bar complex

Definition 2.1 (Fulton–MacPherson realization of the ordered bar;). Let $\mathcal{A}$ be a strictly augmented chiral algebra on a smooth curve X , let $\tilde{\mathcal{A}} = \ker(\varepsilon)$, and let $j_n: \text{Conf}_n(X) \hookrightarrow X^n$ and $\pi_n: \overline{\text{FM}}_n(X) \rightarrow X^n$ be the open inclusion and blowdown map. The enhanced bar $B_X(\mathcal{A})$ is formed first in $D(\text{Ran} X)$. The Fulton–MacPherson jet-realization package $H_{\text{FM}}^{\text{jet}}$ identifies its arity- n chain model with

$$B_{X,n}^{\text{ord}}(\mathcal{A}) \simeq R\Gamma\left(\overline{\text{FM}}_n(X), \text{DR}_{\log}\left(\pi_n^! j_{n,*} j_n^*(s^{-1} \tilde{\mathcal{A}})^{\boxtimes n}\right)\right),$$

with the full logarithmic de Rham complex and its total grading. The summand with $\Omega_{\log}^{n-1}$ is one graded piece of this complex. The ordered object precedes passage to homotopy Σ_n -coinvariants. Here s^{-1} is desuspension and $|d| = +1$.

The bar differential decomposes into three pieces:

$$d_{\overline{B}} = d_{\text{int}} + d_{\text{res}} + d_{\text{dR}}.$$

The internal differential d_{int} acts on each tensor factor of $\mathcal{A}$. The de Rham differential d_{dR} acts on logarithmic forms on $\overline{\text{FM}}_n(X)$. The collision differential combines two pieces of geometry. The Poincaré residue of η_{ij} records the boundary incidence, while the normal-jet filtration of $j_* j^*(\mathcal{A} \boxtimes \mathcal{A})$ records the OPE pole order. On the q -th principal-part graded piece, $q \geq 1$, their contraction is

$$\text{gr}_{D_{ij}}^q d_{\text{res}}^{(ij)}(a_1 \otimes \dots \otimes a_n \otimes (z_i - z_j)^{-q} \otimes \eta_{ij} \wedge \omega') = \pm a_{i(q-1)} a_j \otimes a_1 \otimes \widehat{\otimes \dots \otimes a_n}^{ij} \otimes \omega'.$$

Here the hat with superscript ij retains every tensor factor except the i -th and j -th. In particular, a pole of order q produces the mode $a_{(q-1)}b$. The jet bundle makes this associated-graded description coordinate-covariant.

2.4 Why $d^2 = 0$: the Borchers selection principle

Theorem 2.2 (Bar nilpotence;). *Under $H_{\text{FM}}^{\text{jet}}$, the Borchers identity (2) implies $d_B^2 = 0$. If the jet realization is faithful on the normal-principal-part filtration, the converse also holds.*

Proof. Expand $(d_{\text{int}} + d_{\text{res}} + d_{\text{dR}})^2$ into nine terms. Three vanish individually: $d_{\text{int}}^2 = 0$ (cochain complex), $d_{\text{dR}}^2 = 0$ (de Rham), and $\{d_{\text{int}}, d_{\text{dR}}\} = 0$ (Leibniz). The remaining terms are controlled by index overlap in the residue operator:

Disjoint indices ($\{i, j\} \cap \{k, \ell\} = \emptyset$): residue operators commute; cancellation is by the Koszul sign rule. The OPE input enters at overlapping collision strata.

One shared index (say $j = k$): three boundary strata meet at the codimension-2 locus $D_{ij\ell}$ of triple collision. The Borchers identity makes the cyclic sum of iterated residues vanish. Arnold's relation contracts the differential-form factor, while the Borchers identity contracts the OPE factor. Faithfulness of the normal-jet realization gives the converse.

Same pair ($\{i, j\} = \{k, \ell\}$): $(\text{Res}_{D_{ij}})^2 = 0$ because the first Poincaré residue removes the unique logarithmic normal factor along D_{ij} . $\square$

Remark 2.3 (Selection table). The bar construction asks the same question at every level of geometry:

Base	Condition for $d^2 = 0$	Selected objects
Point	Jacobi identity	Lie algebras
Curve X	Borchers identity	Chiral algebras
X over $\overline{\mathcal{M}}_{g,n}$ in the stable range	+ period corrections	Modular Koszul algebras

Each row is forced by the geometry of the next.

2.5 Two coproducts

The ordered bar carries the *deconcatenation coproduct* (1): $n + 1$ terms, coassociative and E_1 , reflecting the linear ordering. The homotopy- Σ_n descent is

$$B_{X,n}^{\Sigma}(\mathcal{A}) := (B_{X,n}^{\text{ord}}(\mathcal{A}))_{h\Sigma_n}.$$

Under the symmetric descent package it inherits the coshuffle coproduct; on the tensor associated graded this coproduct has 2^n summands and is cocommutative. The ordered bar is the level-2 object before symmetrisation; the symmetric bar is the completed coinvariant descent. In the braided case this is the L_R -twisted descent, and the formal conductor fibre records the non-coinvariant R -matrix data.

2.6 From genus 0 to all genera

At genus $g \geq 1$, the prime form globalizes the local difference. A normalized logarithmic kernel is $d \log E(z_1, z_2)$, where $E(z, w)$ is the prime form on a Riemann surface of genus g (a section of $K^{-1/2} \boxtimes$

$K^{-1/2}$, weight 1). The second-kind bar coalgebra carries a curvature functional, and its curved cobar presentation carries an element $m_o^{(g)}$ whose scalar realization is

$$\mathrm{tr}_{\mathrm{diag}}(m_o^{(g)}) = \kappa(\mathcal{A}) \cdot \omega_g,$$

after scalar diagonal projection of the matrix-valued ordered curvature residue $r_{\mathcal{A}}^{\mathrm{coll}}(z) \cdot \omega_g \in \mathrm{End}_{\mathcal{A}}(2) \otimes H^{1,1}(C_g)$. The raw chain identity is $m_1^2 = [m_o^{(g)}, -]$.

Under the modular Maurer–Cartan package, the *total* bar differential $D_{\mathcal{A}} = \sum_{g \geq o} \hbar^g d_{\mathcal{A}}^{(g)}$ incorporates the curvature and satisfies $D_{\mathcal{A}}^2 = o$ in the completed ambient.

Definition 2.4 (Modular characteristic). The *modular characteristic* $\kappa(\mathcal{A})$ is the normalized degree-two collision trace. In scalar or abelian cases it equals the value of $\tau_{\mathcal{A}}$ on the collision kernel. For nonabelian affine Kac–Moody algebras in the trace-form normalization,

$$\kappa(V_k(\mathfrak{g})) = \tau_{V_k(\mathfrak{g})}(r^{\mathrm{coll}}) + \frac{\dim(\mathfrak{g})}{2} = \frac{\dim(\mathfrak{g})(k + h^{\vee})}{2h^{\vee}},$$

where the second summand is the Sugawara simple-pole contribution.

Definition 2.5 (Maurer–Cartan element). The *Maurer–Cartan element* is $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_o \in \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$, the positive-genus part of the bar differential. Since $D_{\mathcal{A}}^2 = o$, it satisfies

$$d_o \Theta_{\mathcal{A}} + \frac{1}{2} [\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = o$$

in the completed modular ambient. The five theorems characterize its reconstruction, duality, trace, and support realizations.

3 Theorem A: Ran reconstruction and Verdier refinement

3.1 Statement

The full bar object carries three canonical exits, each with its own target and hypothesis package:

$$\begin{aligned} R_X(\mathcal{A}) &:= \Omega_X B_X(\mathcal{A}) & \xrightarrow{\varepsilon_{\mathcal{A}}} & \mathcal{A} & \quad (\text{Ran reconstruction}), \\ \mathcal{A}^i & & \xrightarrow{q_{\mathcal{A}}} & B_X(\mathcal{A}) & \quad (\text{quadratic compression}), \\ K_X(\mathcal{A}) &:= \mathbb{D}_{\mathrm{Ran}} B_X(\mathcal{A}) & & & \quad (\text{Verdier–Koszul algebra}). \end{aligned}$$

Reconstruction returns the source algebra. Quadratic compression compares a presentation-dependent coalgebra with the full bar. Verdier duality transports the full bar coalgebra to an algebra.

Definition 3.1 (The Ran ambient and four hypothesis packages). Let $C_X = (D(\mathrm{Ran} X), \otimes^{\mathrm{ch}})$. The proof of Francis–Gaitsgory’s Theorem 5.1.1 makes this chiral monoidal category pro-nilpotent, and their Proposition 4.1.2 applies to the reduced associative operad. The package H_{fact} asserts closure of the chosen associative factorization subcategories under enhanced bar, enhanced cobar, collision base change, units, and completed inverse limits. The package $H_{\mathrm{CL}}(\mathcal{A}, \mathcal{A}^i, \tau_i)$ supplies a connected positive-weight quadratic coalgebra $\mathcal{A}^i$, its canonical twisting morphism $\tau_i: \mathcal{A}^i \rightarrow \mathcal{A}$, finite-window comparison, and acyclicity of the associated twisted tensor products. The package H_{conv} realizes the completed chain

model through exhaustive complete filtrations and cohomologically Mittag–Leffler inverse systems. The package H_{VD} places $B_X(\mathcal{A})$ in a constructible or holonomic dualizable subcategory and supplies the Kunnet and base-change maps used by Verdier duality.

Theorem 3.2 (Theorem A: typed bar–cobar reconstruction;). *TYPE SIGNATURE. Quadrant: Open. Presentation: chain-level ∞ -category. Level: $1 \leftrightarrow 2$. Ambient: the pro-nilpotent Francis–Gaitsgory chiral Ran category with the reduced associative operad.*

For the reduced associative operad in $C_X = (D(\text{Ran}X), \otimes^{\text{ch}})$, enhanced bar and cobar are mutually inverse equivalences [7, Proposition 4.1.2, together with the pro-nilpotence argument in the proof of Theorem 5.1.1]. For an augmented algebra $\mathcal{A}$ and a divided-power conilpotent coalgebra C , the unit and counit are equivalences

$$C \xrightarrow{\sim} B_X \Omega_X(C), \quad \Omega_X B_X(\mathcal{A}) \xrightarrow{\sim} \mathcal{A}.$$

Conjecture 3.3 (Factorization restriction;). *Under H_{fact} enhanced bar and cobar preserve the factorization equivalences and collision maps, and the ambient Francis–Gaitsgory equivalence restricts to the associative factorization subcategories used in this volume.*

Proposition 3.4 (Quadratic compression;). *Under $H_{\text{CL}}(\mathcal{A}, \mathcal{A}^i, \tau_i)$, the canonical map*

$$q_{\mathcal{A}}: \mathcal{A}^i \xrightarrow{\sim} B_X(\mathcal{A})$$

is a quasi-isomorphism. Local finite-dimensionality forms the quadratic dual algebra $\mathcal{A}^! = (\mathcal{A}^i)^\vee$.

Proposition 3.5 (Verdier refinement;). *Under H_{VD} , Verdier duality transports the coalgebra $B_X(\mathcal{A})$ to the algebra*

$$K_X(\mathcal{A}) := \mathbb{D}_{\text{Ran}} B_X(\mathcal{A}),$$

and biduality gives $B_X(\mathcal{A}) \xrightarrow{\sim} \mathbb{D}_{\text{Ran}} K_X(\mathcal{A})$. Together with quadratic compression, this identifies $K_X(\mathcal{A})$ with the continuous Verdier realization of $\mathcal{A}^!$.

Definition 3.6 (The KSDual fixed locus;). A point belongs to KSDual when it carries a chosen equivalence $\sigma_{\mathcal{A}}: \mathcal{A} \xrightarrow{\sim} K_X(\mathcal{A})$ compatible with the evaluation and coevaluation maps. This is the $\mathbb{Z}/2$ -fixed sublocus of the specified duality correspondence.

Remark 3.7 (Reconstruction, compression, and duality). The counit $R_X(\mathcal{A}) \rightarrow \mathcal{A}$ is the reconstruction morphism. The map $q_{\mathcal{A}}$ compares the quadratic coalgebra with the full bar. The algebra $K_X(\mathcal{A})$ is the Verdier image of the full bar. A comparison $R_X(\mathcal{A}) \simeq K_X(\mathcal{A})$ is precisely reconstruction composed with a self-duality equivalence $\sigma_{\mathcal{A}}: \mathcal{A} \xrightarrow{\sim} K_X(\mathcal{A})$.

Remark 3.8 (Completed realization for unbounded weights). An unbounded conformal-weight grading is represented by its inverse system of finite-weight windows. The package H_{conv} requires cohomological Mittag–Leffler transition maps and the comparison with the derived inverse limit. This completed realization is the ambient used for the Virasoro and principal $\mathcal{W}$ charts.

3.2 Proof

Francis–Gaitsgory prove enhanced operadic bar–cobar equivalence in the pro-nilpotent Ran category. The maps in H_{fact} carry that equivalence to the chosen factorization subcategories. The twisting-morphism fundamental theorem identifies acyclicity of $\mathcal{A} \otimes_{\tau} \mathcal{A}^i$ with the quasi-isomorphism $q_{\mathcal{A}}$. Finally, H_{VD} transports the coalgebra structure through Verdier duality and supplies biduality.

Example 3.9 (Square-zero local model). Let $\mathcal{A} = k[\epsilon]/(\epsilon^2)$ with its standard augmentation. The reduced bar has one basis word $(s^{-1}\epsilon)^{\otimes r}$ in every positive length and zero bar differential. Its deconcatenation coalgebra is the quadratic coalgebra $\mathcal{A}^i$, so $q_{\mathcal{A}}$ is an isomorphism. The cobar counit $\Omega B(\mathcal{A}) \rightarrow \mathcal{A}$ is the standard free resolution. Finite-dimensional duality on each length gives the completed quadratic-dual algebra. This example exhibits reconstruction, quadratic compression, and duality as three distinct maps.

4 Theorem B: quadratic Koszul recognition

4.1 Statement

Theorem A reconstructs an algebra from its full bar in the Ran ambient. Theorem B characterizes the quadratic coalgebras that represent this full bar object.

Theorem 4.1 (Theorem B: quadratic comparison criterion;). *TYPE SIGNATURE. Quadrant: Open. Presentation: chain-level in the associative factorization category. Level: 1 $\leftrightarrow$ 2. Hypothesis package: $H_{\text{CL}}(\mathcal{A}, \mathcal{A}^i, \tau_1)$.*

Let $\mathcal{A} = T_X(V)/(R)$ be a connected positive-weight quadratic presentation and let

$$\mathcal{A}^i = C_X(s^{-1}V, s^{-2}R), \quad \tau_1: \mathcal{A}^i \longrightarrow \mathcal{A}$$

be its quadratic coalgebra and canonical twisting morphism. The package H_{CL} supplies factorization descent, compatible finite-weight windows, convergence of the bar filtration, and the twisted tensor products in the selected completed ambient. Under this package, the following conditions are equivalent:

- (B1) *the canonical coalgebra map $q_{\mathcal{A}}: \mathcal{A}^i \rightarrow B_X(\mathcal{A})$ is a quasi-isomorphism;*
- (B2) *the cobar comparison $\Omega_X(\mathcal{A}^i) \rightarrow \mathcal{A}$ is a quasi-isomorphism;*
- (B3) *the left and right twisted tensor products associated with τ_1 are acyclic in positive weight.*

The additional package $H_{\text{PBW}}^{\text{det}}$ consists of strict filtration-preserving comparison maps, an associated-graded quadratic identification, a splitting that detects cohomology, and a specified E_2 page. Under $H_{\text{CL}} \cup H_{\text{PBW}}^{\text{det}}$, the preceding conditions are also equivalent to

- (B4) *the PBW/Li-bar spectral sequence collapses at E_2 , and E_{∞} is supported on the Koszul diagonal.*

Whenever these conditions hold, local finite-dimensionality forms the quadratic dual algebra $\mathcal{A}^! = (\mathcal{A}^i)^{\vee}$, and the Verdier package of Theorem A compares it with $K_X(\mathcal{A})$.

4.2 Proof architecture

The twisting-morphism fundamental theorem identifies the acyclicity of the two twisted tensor products with the quasi-isomorphisms in conditions (B1) and (B2). The strict filtration and detecting splitting in $H_{\text{PBW}}^{\text{det}}$ identify the associated graded cone of $q_{\mathcal{A}}$ with the off-diagonal part of the quadratic Koszul complex. Collapse at E_2 and diagonal support identify the cone with an acyclic complex. The same argument in each finite-weight window, followed by the cohomological Mittag-Leffler comparison, gives the completed statement.

Example 4.2 (Polynomial algebra). For $\mathcal{A} = k[x]$ with x in positive weight, the quadratic coalgebra is the exterior coalgebra on $s^{-1}x$. The map $\mathcal{A}^i \rightarrow B(\mathcal{A})$ includes the standard Koszul cycles, and the

Koszul contraction gives a quasi-isomorphism. Equivalently, $\mathcal{A} \otimes_{\tau} \mathcal{A}^i$ is the two-term Koszul resolution of the augmentation module. This is the first non-trivial instance of all four conditions.

Remark 4.3 (Recognition boundary). For a family whose cone $\text{Cone}(q_{\mathcal{A}})$ carries cohomology, the quadratic coalgebra records a proper compression of the full bar. The full Ran count of Theorem A and the quadratic comparison of Theorem B retain their separate domains. For an unbounded conformal-weight family, a comparison between direct-sum and completed product totalizations additionally requires finite-window contractions compatible with the transition maps.

5 Theorem C: the 5×5 stratification atlas

5.1 Five measurements

The symbol κ occurs in five native categories. We therefore write

$$\kappa(\mathcal{A}) := \left(\kappa_{\text{cat}}(\mathcal{A}), \kappa_{\text{ch}}^{\text{Hodge}}(\mathcal{A}), \kappa_{\text{ch}}^{\text{Heis}}(\mathcal{A}), \kappa_{\text{BKM}}(\mathcal{A}), \kappa_{\text{fiber}}(\mathcal{A}) \right).$$

These coordinates are respectively categorical, Hodge-graded chiral, Heisenberg-normalized chiral, Borchers, and topological. An equality between two coordinates is represented by a comparison map between their source objects.

5.2 Three comparison tiers

Theorem C has three typed tiers.

C1. The strict flat fibre-centre comparison is the map

$$Z(\Omega_X B_X(\mathcal{A})) \xrightarrow{Z(\varepsilon_{\mathcal{A}})} Z(\mathcal{A}) = H^{\circ} C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$$

on a chirally Koszul datum.

- C2. A represented Verdier involution, perfectness, and a nondegenerate anti-invariant pairing give the homotopy eigenspace decomposition and its scalar trace shadow.
- C3. A shifted-symplectic/BV package upgrades the represented eigenspaces to a Lagrangian intersection in the global deformation space. Its data include the BV Laplacian, the QME anomaly class, and a chosen null-homotopy of that class whenever QME anomaly cancellation is asserted.

The scalar relation belongs to the trace realization of C1 together with Theorem D. The shifted-symplectic assertion belongs to C2 and carries its own construction.

Definition 5.1 (Family comparison package for Theorem C;). For a family $\mathcal{F}$, the package $H_C(\mathcal{F})$ consists of:

- (C1) a source object and trace functional for every displayed coordinate of κ ;
- (C2) a represented Verdier involution on the selected perfect ambient and its evaluation-coevaluation maps;
- (C3) an anti-invariant pairing whenever a complementarity decomposition is asserted;
- (C4) a BV and shifted-symplectic package whenever the C2 enhancement is asserted;
- (C5) cross-quadrant realization maps for every Open-CY equality.

5.3 Statement

Theorem 5.2 (Theorem C: census-scoped 5×5 stratification;). *TYPE SIGNATURE. Quadrants: Open, with a conditional Open-CY row. Presentations: categorical, chain, scalar trace, and topological. Levels: 0–5. Hypothesis package: $H_C(\mathcal{F})$.*

For every family carrying $H_C(\mathcal{F})$, the five measurements form the row $\kappa(\mathcal{F})$. The finite atlas uses the family labels

$$G, \quad L, \quad C, \quad M, \quad B.$$

The first four label the selected Vol I standard-family census. The fifth is the conditional $K_3 \times E$ comparison row. Under its realization package H_B , that row is

$$\kappa(B) = (0, 0, 3, 5, 24).$$

Here $\text{wt}(\Delta_5) = 5$ [14] and $e_{\text{top}}(K_3) = 24$ [17] are exact native calculations. In the stated normalizations the categorical and Hodge entries are the native zeros. The Open-quadrant Heisenberg entry 3 is transported by the comparison $\eta_{K_3, E}$ in H_B . The holomorphic Euler characteristic $\chi(\mathcal{O}_{K_3}) = 2$ remains a distinct categorical invariant.

Suppose the represented Verdier package gives a perfect involution ι_{VD} on a genus- g ambient complex $\mathbf{C}_g(\mathcal{A})$. Then its idempotents

$$e_{\pm} = \frac{1}{2}(\text{id} \pm \iota_{\text{VD}})$$

give the homotopy decomposition

$$\mathbf{C}_g(\mathcal{A}) \simeq \text{im}(e_+) \oplus \text{im}(e_-). \quad (3)$$

When the anti-invariant perfect pairing identifies these two images with the chart and dual-chart sectors, cohomology gives

$$H^{\bullet}(\mathbf{C}_g(\mathcal{A})) \cong Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^{\dagger}). \quad (4)$$

A scalar relation

$$\kappa(\mathcal{A}) + \kappa(\mathcal{A}^{\dagger}) = K_{\mathcal{F}} \quad (5)$$

is the trace of a represented family comparison. Its constant $K_{\mathcal{F}}$ belongs to that family datum.

Finally, the C2 package upgrades (3) to the corresponding shifted-symplectic Lagrangian statement.

Proof. The five coordinates are the trace functionals in $H_C(\mathcal{F})$. The idempotent identities $e_{\pm}^2 = e_{\pm}$, $e_+e_- = 0$, and $e_+ + e_- = \text{id}$ give (3). Perfectness and the represented pairing identify the two images and give (4). Applying the scalar trace to the represented Verdier comparison gives (5). The final clause is the shifted-symplectic structure supplied by the C2 package. $\square$

Proposition 5.3 (Worked scalar pairings;). *A represented trace comparison gives the familiar scalar formulas*

$$\kappa(\beta\gamma_{\lambda}) + \kappa(bc_{\lambda}) = 0, \quad \kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = 13.$$

The first follows from $\kappa(\beta\gamma_{\lambda}) = 6\lambda^2 - 6\lambda + 1$ and $\kappa(bc_{\lambda}) = -\kappa(\beta\gamma_{\lambda})$. The second follows from $\kappa(\text{Vir}_c) = c/2$. At $c = 13$, the two Virasoro scalar coordinates agree.

Remark 5.4 (KSDual fixed locus). A KSDual point carries a chosen equivalence

$$\sigma_{\mathcal{A}}: \mathcal{A} \xrightarrow{\sim} K_X(\mathcal{A})$$

compatible with the represented pairing. Its scalar coordinate is then fixed by the induced trace comparison. The object-level equivalence and its scalar trace occupy successive levels of the tower.

6 Theorem D: the modular characteristic hierarchy

6.1 The four typed objects

The genus expansion carries four successive realizations of the modular anomaly. Their ambient groups determine the meaning of every formula.

Definition 6.1 (Deformation class, virtual object, Hodge shadow, and graph trace;). For $g \geq 2$, let

$$\text{Def}_g(\mathcal{A}) := R\Gamma\left(\overline{\mathcal{M}}_g, \mathcal{E}nd(\tilde{B}_{\text{flat}}^{(g)}(\mathcal{A}))\right).$$

The native Maurer–Cartan obstruction is

$$\text{Obs}_g^{\text{def}}(\mathcal{A}) \in H^2(\text{Def}_g(\mathcal{A})).$$

A perfect scalar realization gives a virtual object

$$\mathfrak{D}_g^K(\mathcal{A}) \in K^0(\overline{\mathcal{M}}_g) \otimes \mathbb{C},$$

whose codimension- g Hodge shadow is

$$\text{obs}_g^{\text{Hdg}}(\mathcal{A}) := \text{ch}_g(\mathfrak{D}_g^K(\mathcal{A})) \in H^{2g}(\overline{\mathcal{M}}_g, \mathbb{C}).$$

A stable-graph functional gives the numerical coefficient $F_g(\mathcal{A}) \in \mathbb{C}$ from the marked stable stack $\overline{\mathcal{M}}_{g,1}$, for $g \geq 1$. At genus one, the pointed deformation complex

$$\text{Def}_{1,1}(\mathcal{A}) := R\Gamma\left(\overline{\mathcal{M}}_{1,1}, \mathcal{E}nd(\tilde{B}_{\text{flat}}^{(1,1)}(\mathcal{A}))\right)$$

is the ambient of $\text{Obs}_{1,1}^{\text{def}}(\mathcal{A}) \in H^2(\text{Def}_{1,1}(\mathcal{A}))$, the source of the normalized one-loop trace.

Lemma 6.2 (Higher-order Maurer–Cartan obstruction class;). *Let L be a complete filtered dg Lie algebra, let $\theta_0 \in L^1$ be a Maurer–Cartan element, and fix $g \geq 2$. Suppose $\theta_{<g}(t) = \theta_0 + \sum_{r=1}^{g-1} t^r \theta_r$ satisfies the Maurer–Cartan equation modulo t^g . Then*

$$o_g := \frac{1}{2} \sum_{\substack{i+j=g \\ 1 \leq i, j \leq g-1}} [\theta_i, \theta_j]$$

is a cocycle for $d_{\theta_0} = d + [\theta_0, -]$. A coefficient θ_g extends the solution through order t^g precisely when $[o_g] = 0$ in $H^2(L, d_{\theta_0})$.

Proof. Write $F(\theta) = d\theta + \frac{1}{2}[\theta, \theta]$. The coefficients of $F(\theta_{<g})$ below order g vanish. The coefficient of t^g in the Bianchi identity $d_\theta F(\theta) = 0$ is therefore $d_{\theta_0} o_g = 0$. Adding $t^g \theta_g$ changes this coefficient to $o_g + d_{\theta_0} \theta_g$, which gives the stated equivalence. $\square$

Remark 6.3 (The genus-one anchor). The sum defining o_g begins at $g = 2$. The genus-one class in Theorem D arises from the normalized one-loop collision trace; the datum H_D^1 identifies that trace with $\kappa(\mathcal{A})\lambda_1$.

6.2 Statement

Theorem 6.4 (Theorem D: modular characteristic hierarchy;). *TYPE SIGNATURE. Quadrant: Open. Presentation: modular deformation complex, perfect K-theory realization, and stable graphs. Levels: $4 \rightarrow 5$. Hypothesis package:*

$$H_D = (H_D^1, H_D^K, H_D^{\text{tr}}, H_D^{\text{graph}}).$$

Let $\mathcal{A} = A_b$ be a cyclic modular Koszul chart algebra and let $\kappa(\mathcal{A})$ be its normalized degree-two collision trace.

(D1) Under $H_D^1(\mathcal{A})$, the normalized genus-one trace is

$$\text{tr}_1 \text{Obs}_{1,1}^{\text{def}}(\mathcal{A}) = \kappa(\mathcal{A})\lambda_1 \in H^2(\overline{\mathcal{M}}_{1,1}). \quad (6)$$

(D2) For $g \geq 2$, under $H_D^K(\mathcal{A}, g)$, the perfect virtual object on $\overline{\mathcal{M}}_g$ is

$$\mathfrak{D}_g^K(\mathcal{A}) = \kappa(\mathcal{A})\lambda_{-1}(\mathbb{E}_g), \quad \lambda_{-1}(\mathbb{E}_g) = \sum_{i=0}^g (-1)^i [\Lambda^i \mathbb{E}_g].$$

Its top Hodge projection is

$$\text{obs}_g^{\text{Hdg}}(\mathcal{A}) = (-1)^g \kappa(\mathcal{A})\lambda_g. \quad (7)$$

The package $H_D^{\text{tr}}(\mathcal{A}, g)$ supplies the comparison from the scalar realization of $\text{Obs}_g^{\text{def}}(\mathcal{A})$ to $\mathfrak{D}_g^K(\mathcal{A})$.

On $\overline{\mathcal{M}}_{1,1}$ the analogous virtual object has degree-one character $-\kappa(\mathcal{A})\lambda_1$. This Hodge character and the positively normalized one-loop trace in (D1) are two realizations connected only by their stated comparison data.

(D3) For $g \geq 1$, under $H_D^{\text{graph}}(\mathcal{A}, g)$, the stable-graph functional is

$$F_g(\mathcal{A}) = \kappa(\mathcal{A})\lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}(\mathcal{A}), \quad \lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g. \quad (8)$$

The scalar-diagonal graph package gives $\delta F_g^{\text{cross}}(\mathcal{A}) = 0$. For a multi-weight chart, the second term collects stable graphs whose edges meet distinct conformal-weight channels.

Remark 6.5 (Scope and normalizations). The genus-one trace in (6) has positive sign by its loop normalization. The Hodge shadow in (7) carries the sign of the alternating exterior power. The graph functional in (8) is numerical. The comparison packages connect these three realizations while preserving their ambient degrees.

6.3 The Chern-root calculation

Proposition 6.6 (Top Hodge projection;). *Let E be a rank- g vector bundle. Then*

$$\mathrm{ch}_g(\lambda_{-1}(E)) = (-1)^g c_g(E).$$

Consequently $H_D^K(\mathcal{A}, g)$ gives $\mathrm{obs}_g^{\mathrm{Hdg}}(\mathcal{A}) = (-1)^g \kappa(\mathcal{A}) \lambda_g$.

Proof. On a splitting space, let $x_1, \dots, x_g$ be the Chern roots of E . Then

$$\mathrm{ch}(\lambda_{-1}(E)) = \prod_{i=1}^g (1 - e^{x_i}) = (-1)^g x_1 \cdots x_g + \text{terms of codimension greater than } g.$$

Since $x_1 \cdots x_g = c_g(E)$, the codimension- g component is the displayed class. $\square$

Remark 6.7 (The comparison problem). A construction of H_D^K consists of a perfect scalar virtual bar object and an equality

$$\beta_g(\mathcal{A}) : \mathfrak{D}_g^K(\mathcal{A}) \xrightarrow{\sim} \kappa(\mathcal{A}) \lambda_{-1}(\mathbb{E}_g)$$

compatible with clutching and modular trace. Grothendieck–Riemann–Roch computes the characteristic class after this comparison is given. Clutching tests its boundary compatibility. The map $\beta_g(\mathcal{A})$ is the load-bearing higher-genus construction.

6.4 The Faber–Pandharipande scalar

Faber–Pandharipande’s λ_g formula [9, Section 0.2] gives

$$\lambda_g^{\mathrm{FP}} = \frac{2^{2g-1} - 1}{2^{2g-1}} \frac{|B_{2g}|}{(2g)!}.$$

Hence the scalar-diagonal graph series is

$$\sum_{g \geq 1} F_g(\mathcal{A}) x^{2g} = \kappa(\mathcal{A}) \left(\frac{x/2}{\sin(x/2)} - 1 \right), \quad \partial F_g^{\mathrm{cross}}(\mathcal{A}) = 0.$$

The first coefficients are

$$\lambda_1^{\mathrm{FP}} = \frac{1}{24}, \quad \lambda_2^{\mathrm{FP}} = \frac{7}{5760}, \quad \lambda_3^{\mathrm{FP}} = \frac{31}{967680}.$$

Example 6.8 (Heisenberg realization;). For the rank-one Heisenberg algebra $[a_m, a_n] = k m \delta_{m+n,0} \mathbf{1}$, one has $\kappa(\mathcal{H}_k) = k$. The packages in Theorem D give

$$\mathrm{tr}_1 \mathrm{Obs}_{1,1}^{\mathrm{def}}(\mathcal{H}_k) = k \lambda_1, \quad \mathfrak{D}_g^K(\mathcal{H}_k) = k \lambda_{-1}(\mathbb{E}_g), \quad g \geq 2, \quad \mathrm{obs}_g^{\mathrm{Hdg}}(\mathcal{H}_k) = (-1)^g k \lambda_g, \quad g \geq 2.$$

On the scalar-diagonal graph lane,

$$\sum_{g \geq 1} F_g(\mathcal{H}_k) x^{2g} = k \left(\frac{x/2}{\sin(x/2)} - 1 \right).$$

7 Theorem H: chiral Hochschild cohomology

7.1 Statement

Theorem D separates the native deformation class, its Hodge shadow, and the numerical graph trace. Theorem H determines the family-specific cohomological support of the level-3 chart centre.

Theorem 7.1 (Theorem H: family-indexed chiral Hochschild support;). *TYPE SIGNATURE. Quadrant: Open. Presentation:*

$$C_{\text{ch}}^{\bullet}(A_b, A_b) := R\text{Hom}_{A_b \otimes A_b^{\text{op}}}(A_b, A_b), \quad \text{ChirHoch}^n(A_b) := H^n C_{\text{ch}}^{\bullet}(A_b, A_b),$$

formed in the completed chiral bimodule category. Level: 3 (derived chiral centre $Z_{\text{ch}}^{\text{der}}(A_b)$, the algebraic closed-sector object in the Beilinson tower). Hypothesis package: the family datum $H_H(\mathcal{A}; S)$.

Fix a finite set $S \subset \mathbb{Z}$. The datum $H_H(\mathcal{A}; S)$ consists of a complete filtered cochain model $Q_{\mathcal{A}}$, a quasi-isomorphism

$$\gamma_{\mathcal{A}}: Q_{\mathcal{A}} \xrightarrow{\sim} C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A}),$$

a complete complex $K_{\mathcal{A}, S}$ whose cochain terms occur in degrees belonging to S , and filtration-continuous maps

$$K_{\mathcal{A}, S} \xrightarrow{\iota_{\mathcal{A}}} Q_{\mathcal{A}} \xrightarrow{p_{\mathcal{A}}} K_{\mathcal{A}, S}, \quad h_{\mathcal{A}}: Q_{\mathcal{A}} \longrightarrow Q_{\mathcal{A}}[-1],$$

satisfying

$$p_{\mathcal{A}} \iota_{\mathcal{A}} = \text{id}_{K_{\mathcal{A}, S}}, \quad dh_{\mathcal{A}} + h_{\mathcal{A}} d = \text{id}_{Q_{\mathcal{A}}} - \iota_{\mathcal{A}} p_{\mathcal{A}}.$$

Whenever a bar model is used, the datum includes the bar-bimodule comparison

$$\Gamma_{\bar{B}}: \text{Conv}(B_X(\mathcal{A}), \mathcal{A}) \xrightarrow{\sim} C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A}).$$

A cyclic pairing may further identify the shifted deformation complex with cyclic coderivations of the completed bar; this is a separate comparison with its suspension recorded explicitly. The datum also supplies convergence of the filtration and, whenever $Q_{\mathcal{A}}$ is first formed on ordered configurations, the ordered-to-symmetric comparison quasi-isomorphism. Then

$$\text{ChirHoch}^n(\mathcal{A}) \cong H^n(K_{\mathcal{A}, S}), \quad \text{Supp ChirHoch}^{\bullet}(\mathcal{A}) \subseteq S. \quad (9)$$

If the displayed cohomology groups are finite-dimensional, their Hilbert polynomial is

$$P_{\mathcal{A}}(t) = \sum_{n \in S} \dim H^n(K_{\mathcal{A}, S}) t^n.$$

Suppose additionally that $\mathcal{A}^{\dagger}$ carries $H_H(\mathcal{A}^{\dagger}; S^{\dagger})$ and that $(\mathcal{A}, \mathcal{A}^{\dagger})$ carries a perfect degree- d chain pairing. This separate duality datum gives

$$\text{ChirHoch}^n(\mathcal{A}) \cong \text{ChirHoch}^{d-n}(\mathcal{A}^{\dagger})^{\vee} \otimes \omega_X, \quad P_{\mathcal{A}}(t) = t^d P_{\mathcal{A}^{\dagger}}(t^{-1}).$$

The geometric pairing determines d .

Remark 7.2 (Three Hochschild theories: chiral, topological, categorical). Three Hochschild theories occupy distinct ambients: chiral, topological, and categorical. *Chiral Hochschild* $\text{ChirHoch}^*(\mathcal{A}) =$

$H^* \mathrm{RHom}_{\mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}}(\mathcal{A}, \mathcal{A})$ lives in the completed chiral bimodule category on X , and $H_H(\mathcal{A}; S)$ computes its family-specific support. For an E_1 ring spectrum R , topological Hochschild homology is $\mathrm{THH}(R) = R \otimes_{R^e}^L R$, equivalently factorization homology on S^1 , while topological Hochschild cohomology is $F_{R^e}(R, R)$. *Categorical Hochschild* $\mathrm{HH}^*(\mathrm{Rep}(\mathcal{A}))$ governs the Drinfeld center and the E_2 structure. Comparison functors between these theories constitute additional mathematical data.

7.2 Family-specific specializations

Proposition 7.3 (Two bounded-cohomology benchmarks;). *Assume that the bounded-to-chart map for the stated vertex algebra is a quasi-isomorphism. The computations of Bakalov–De Sole–Kac then give the following chart cohomology.*

(i) *For the rank-one even free superboson $B_{\mathfrak{b}}$,*

$$(\dim \mathrm{ChirHoch}^n(\mathcal{A}_{B_{\mathfrak{b}}}))_{n \geq 0} = (2, 1, 0, 0, \dots).$$

(ii) *For the Virasoro vertex algebra Vir_c ,*

$$\dim \mathrm{ChirHoch}^n(\mathcal{A}_{\mathrm{Vir}_c}) = \begin{cases} 1, & n \in \{0, 2, 3\}, \\ 0, & n \in \mathbb{Z}_{\geq 0} \setminus \{0, 2, 3\}. \end{cases}$$

These vectors are the bounded calculations of [4, Theorems 7.4 and 7.2]; the assumed comparison carries them to the chart complex. Affine, principal $\mathcal{W}$, lattice, and critical-level families acquire their support from their respective data $H_H(\mathcal{A}; S_{\mathcal{A}})$. At affine critical level the Feigin–Frenkel centre supplies a degree-zero subspace modeled by $\mathrm{Fun}(\mathrm{Op}_{\mathfrak{g}^{\vee}}(D))$. The complete critical-level datum H_H must additionally determine every other cohomological degree in the completed chart model.

7.3 Proof architecture

The identities in the datum $H_H(\mathcal{A}; S)$ exhibit $K_{\mathcal{A}, S}$ and $Q_{\mathcal{A}}$ as homotopy-equivalent cochain complexes. Composition with $\gamma_{\mathcal{A}}$ proves (9). A finite-window construction establishes these identities stratum by stratum on the Fulton–MacPherson compactification, verifies compatibility with the bar faces and collision incidences, passes through the ordered-to-symmetric comparison, and then takes the derived inverse limit. PBW filtrations and quadratic Koszul complexes furnish one construction method; the resulting homotopy and convergence identities carry the theorem.

The perfect chain pairing with the dual chart is an additional geometric structure. Its degree gives the reflection $n \leftrightarrow d - n$, and finite-dimensionality then gives the palindromic Hilbert identity.

8 The dependency graph among Theorems A, B, C, D, and H

The five theorems form a typed dependency graph. The graph has two principal branches: reconstruction and duality at levels 1–3, and modular trace at levels 4–5.

$$\begin{array}{ccccc}
\mathcal{A}^i & \xrightarrow{q_{\mathcal{A}}} & B_X(\mathcal{A}) & \xrightarrow{\Omega_X} & \mathcal{A} \\
& & \downarrow \mathbb{D}_{\text{Ran}} & & \downarrow Z_{\text{ch}}^{\text{der}} \\
& & K_X(\mathcal{A}) & & C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})
\end{array}$$

$$K_{\mathcal{A}, S} \xrightleftharpoons[p_{\mathcal{A}}]{\iota_{\mathcal{A}}} Q_{\mathcal{A}} \xrightarrow{\gamma_{\mathcal{A}}} C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$$

The upper two rows display quadratic compression, reconstruction, Verdier duality, and the derived centre. The third row is the strong deformation retract and chart comparison of Theorem H. Theorem D has the separate typed diagram

$$\begin{array}{ccccc}
\text{Obs}_g^{\text{def}}(\mathcal{A}) & \xrightarrow{H_D^{\text{tr}}} & \mathfrak{O}_g^K(\mathcal{A}) & \xrightarrow{\text{ch}_g} & \text{obs}_g^{\text{Hdg}}(\mathcal{A}) \\
& \searrow H_D^{\text{graph}} & & & \\
& & F_g(\mathcal{A}) & &
\end{array}$$

- (i) The Ran bar–cobar equivalence of Theorem A constructs $R_X(\mathcal{A}) = \Omega_X B_X(\mathcal{A}) \simeq \mathcal{A}$. The Verdier package constructs the separate algebra $K_X(\mathcal{A}) = \mathbb{D}_{\text{Ran}} B_X(\mathcal{A})$.
- (ii) Theorem B compares the quadratic coalgebra $\mathcal{A}^i$ with the full bar through $q_{\mathcal{A}}: \mathcal{A}^i \rightarrow B_X(\mathcal{A})$. Its input is H_{CL} , and its conclusion is the equivalence of the three intrinsic twisting conditions. The fourth, spectral-sequence condition joins them under $H_{\text{PBW}}^{\text{det}}$.
- (iii) Theorem C organizes the five scalar measurements and their family strata. A represented Verdier comparison supplies any relation between the $\mathcal{A}$ - and $\mathcal{A}^!$ -rows.
- (iv) Theorem D starts from the cyclic modular trace. Its packages compare the native deformation class with a virtual K -object, take the signed Hodge projection, and evaluate the stable-graph functional.
- (v) Theorem H starts from the complete chiral Hochschild chart complex. The family datum $H_H(\mathcal{A}; S)$ gives a strong deformation retract onto a complex supported in S .

Remark 8.1 (Primitive anchor and comparison arrows). The full bar differential and its Maurer–Cartan element are defined before the five theorem packages. Theorem A applies the Ran bar–cobar equivalence. Theorem B adds a quadratic model. Theorem C adds family-indexed scalar measurements. Theorem D adds modular trace and Hodge realization. Theorem H adds a complete family support model. Each comparison arrow therefore introduces a new mathematical datum with a specified source and target.

8.1 One new input per theorem

Theorem	Output	Input	Status
A	$R_X(\mathcal{A}) \simeq \mathcal{A}, K_X(\mathcal{A})$	Ran ambient; $H_{\text{fact}}, H_{\text{conv}}, H_{\text{VD}}$	published / conjectural / conditional
B	$q_{\mathcal{A}}: \mathcal{A}^i \xrightarrow{\sim} B_X(\mathcal{A})$	$H_{\text{CL}}, H_{\text{PBW}}^{\text{det}}$	conditional
C	five family measurements	represented comparison data	family-indexed
D	$\text{Obs}_g^{\text{def}}, \mathfrak{D}_g^K, \text{obs}_g^{\text{Hdg}}, F_g$	$H_D^i, H_D^K, H_D^{\text{tr}}, H_D^{\text{graph}}$	conditional
H	$\text{Supp ChirHoch}^\bullet(\mathcal{A}) \subseteq S$	$H_H(\mathcal{A}; S)$	conditional

9 The Heisenberg chart: the first complete local calculation

9.1 Exact local algebra

Let $\mathcal{H}_k$ be the rank-one Heisenberg vertex algebra generated by a weight-one field J . Its defining operator product and λ -bracket are

$$J(z)J(w) \sim \frac{k}{(z-w)^2}, \quad [J_\lambda J] = k\lambda.$$

Equivalently,

$$[J_m, J_n] = k m \delta_{m+n, 0} \mathbf{1}.$$

The normalized scalar collision seed is therefore

$$R_2^{\text{sc}}(\mathcal{H}_k) = \kappa(\mathcal{H}_k) = k.$$

This equality is the exact local computation. The strong-generator space is one-dimensional, while the state space contains the full descendant tower. At $k = 0$, the vacuum projection is an ordinary augmentation. At $k \neq 0$, the central term $k\lambda$ makes the vacuum projection a curved splitting.

9.2 Theorem A: full Ran reconstruction

The curved second-kind chiral bar coalgebra is

$$B_X^{\Pi}(\mathcal{H}_k) = \left(T_X^c(s^{-1}\overline{\mathcal{H}}_k), d_B, h_k \right), \quad h_k: B_X^{\Pi}(\mathcal{H}_k) \longrightarrow \omega_X[2],$$

where $\overline{\mathcal{H}}_k$ is the chosen vacuum complement and includes the descendants of J , and $d_B = d_{\text{int}} + d_{\text{coll}}$ is assembled from all collision strata. The curvature functional h_k is supported on length two and represents the central collision

$$[J_\lambda J] = k\lambda.$$

In the sign convention of this chapter,

$$d_B^2 = (h_k \otimes \text{id} - \text{id} \otimes h_k)\Delta.$$

Its curved cobar algebra $\Omega_X^\Pi B_X^\Pi(\mathcal{H}_k)$ carries $m_o = -k\omega$. For $k = o$, the curvature functional vanishes and the Ran theorem of Francis–Gaitsgory gives

$$R_X(\mathcal{H}_o) = \Omega_X B_X(\mathcal{H}_o) \xrightarrow{\sim} \mathcal{H}_o$$

in the pro-nilpotent Ran ambient. For $k \neq o$, a completed curved bar–cobar comparison belongs to the second-kind package $H_{\text{curv}}(\mathcal{H}_k)$; its conclusion is a coderived equivalence represented by the curved counit.

Under the curved Verdier package, duality gives the separate algebra

$$K_X^\Pi(\mathcal{H}_k) := \mathbb{D}_{\text{Ran}} B_X^\Pi(\mathcal{H}_k).$$

A strict presentation of $K_X^\Pi(\mathcal{H}_k)$ as a curved Heisenberg dual requires the quadratic comparison, local finite duality, and a formality map.

9.3 Theorem B: the finite-window comparison

Let $\mathcal{H}_k^{i,II}$ be the curved quadratic coalgebra determined by the strong generator, the bracket, and the induced length-two curvature functional h_k^i . The Heisenberg instance of Theorem B is the curvature-preserving map

$$q_{\mathcal{H}_k}^\Pi : \mathcal{H}_k^{i,II} \longrightarrow B_X^\Pi(\mathcal{H}_k).$$

The datum $H_{\text{CL}}^\Pi(\mathcal{H}_k, \mathcal{H}_k^{i,II}, \tau)$ consists of explicit contractions on every finite conformal-weight window, compatibility with collision incidences and bar faces, and the Mittag–Leffler passage to the completed chart. Once this datum is constructed, it gives the curved quadratic comparison and the second-kind cobar counit. At $k = o$, these specialize to the ordinary quasi-isomorphisms of Theorem B. Thus the one-line OPE supplies the local seed, and the finite-window contraction supplies the global chiral theorem.

9.4 Theorems C and D: scalar and Hodge realizations

The row-C input furnished directly by the OPE is $\kappa(\mathcal{H}_k) = k$. A represented curved Verdier trace comparison may identify the dual scalar with $-k$; on that comparison surface,

$$\kappa(\mathcal{H}_k) + \kappa(K_X^\Pi(\mathcal{H}_k)) = o.$$

This is a relation between represented trace functionals.

Under the packages of Theorem D,

$$\text{tr}_I \text{Obs}_{i,I}^{\text{def}}(\mathcal{H}_k) = k\lambda_1, \quad \mathfrak{D}_g^K(\mathcal{H}_k) = k\lambda_{-1}(\mathbb{E}_g), \quad g \geq 2, \quad \text{obs}_g^{\text{Hdg}}(\mathcal{H}_k) = (-1)^g k\lambda_g, \quad g \geq 2.$$

On the scalar-diagonal stable-graph lane,

$$F_g(\mathcal{H}_k) = k\lambda_g^{\text{FP}}, \quad \sum_{g \geq 1} F_g(\mathcal{H}_k) x^{2g} = k \left(\frac{x/2}{\sin(x/2)} - 1 \right).$$

9.5 Theorem H: bounded benchmark and chart transport

For the rank-one even free superboson $B_{\mathfrak{h}}$, Bakalov–De Sole–Kac prove

$$\dim H_{\text{ch},b}^n(B_{\mathfrak{h}}) = \begin{cases} 2, & n = 0, \\ 1, & n = 1, \\ 0, & n \geq 2, \end{cases} \quad [4, \text{Theorem 7.4}].$$

Fix $k \in \mathbb{C}^\times$ and choose $u \in \mathbb{C}^\times$ with $u^2 = k$. The rescaling $\rho_u: B_{\mathfrak{h}} \xrightarrow{\sim} \mathcal{H}_k$, $a \mapsto u^{-1}J$, transports the normalized bracket to $[J_\lambda J] = k\lambda$. A bounded-to-chart quasi-isomorphism

$$\chi_{\mathcal{H}_k}^{\text{bd}}: C_{\text{ch},b}^\bullet(B_{\mathfrak{h}}) \xrightarrow{C^\bullet(\rho_u)} C_{\text{ch},b}^\bullet(\mathcal{H}_k) \xrightarrow{\sim} C_{\text{ch}}^\bullet(\mathcal{A}_{\mathcal{H}_k}, \mathcal{A}_{\mathcal{H}_k})$$

transports this vector to the chart complex. In the language of Theorem H, that comparison supplies the family support datum with $S_{\mathcal{H}} = \{0, 1\}$ and cohomology vector $(2, 1, 0, \dots)$. The fiber $k = 0$ is the commutative current algebra and carries its own family datum.

Remark 9.1 (Abelian Chern–Simons comparison). For the standard chiral boundary condition of abelian Chern–Simons theory at level k , the boundary current algebra is $\mathcal{H}_k$. The equality $\kappa(\mathcal{H}_k) = k$ matches the OPE coefficient with the gauge-theory level. A comparison between the stable-graph functional above and a perturbative Chern–Simons amplitude is an additional open–closed map.

9.6 Summary of the worked case

Theorem	Heisenberg statement	Load-bearing datum
A	ordinary Ran counit at $k = 0$; curved coderived counit at $k \neq 0$	Ran ambient; $H_{\text{curv}}, H_{\text{conv}}, H_{\text{VD}}^{\text{II}}$
B	$q_{\mathcal{H}_k}^{\text{II}}: \mathcal{H}_k^{\text{I,II}} \rightarrow B_X^{\text{II}}(\mathcal{H}_k)$	finite-window $H_{\text{CL}}^{\text{II}}$
C	$\kappa(\mathcal{H}_k) = k$	represented dual trace for the paired row
D	$\text{obs}_g^{\text{Hdg}} = (-1)^g k \lambda_g$	H_D^K
H	bounded vector $(2, 1, 0, \dots)$	bounded-to-chart comparison

The exact local input is the bracket $[J_\lambda J] = k\lambda$. Every global conclusion records the map that carries this local input through configurations, completion, duality, Hodge realization, or cohomological comparison.

10 The standard-family atlas: source data and comparison status

The standard-family atlas is finite. Each row begins with an exact local calculation and acquires its global entries through named comparison maps. The atlas is the finite census of constructed and conditional rows.

10.1 Local calculations before global transport

The local-datum and benchmark columns record calculations in their stated normalizations. The comparison column names the map into the chiral bar, Hodge, Hochschild, or cross-volume ambient; the final column records the resulting epistemic status.

Table 1: Source data, global comparison, and epistemic status.

Family	Local datum	Benchmark	Global comparison	Status
Heisenberg $\mathcal{H}_k$	$[J_\lambda J] = k\lambda, \kappa = k$	bounded superboson vector $(2, 1, 0, \dots)$	H_{CL}, H_D , bounded-to-chart map	OPE and κ proved; transport conditional
Affine $V_k(\mathfrak{g})$	$[J_\lambda^a J^b] = [J^a, J^b] + k\lambda(a, b)$	$\kappa = \frac{(k + h^\vee) \dim \mathfrak{g}}{2h^\vee}$	quadratic contraction, represented trace, family H_H	κ proved in trace normalization; transport conditional
$\beta\gamma_\lambda, bc_\lambda$	free-field Wick algebra	$\kappa_{\beta\gamma} = 6\lambda^2 - 6\lambda + 1, \kappa_{bc} = -\kappa_{\beta\gamma}$	collision-residue descent and family support datum	Wick values computed; transport conditional
Virasoro Vir_c	$T(z)T(w)$ OPE, $\kappa = c/2$	level-four Gram coefficient $10/[c(5c + 22)], c(5c + 22) \neq 0$; bounded support $\{0, 2, 3\}$	H_{res} , bounded-to-chart map, H_D	Gram seed exact on domain; chart support conditional
Principal $\mathcal{W}_N$	strong weights $2, \dots, N$	$\kappa = c(H_N - 1)$	mixed-channel graph functional and family H_H	value conditional on DS/bar trace comparison
$K_3 \times E$ comparison row	$e_{\text{top}}(K_3) = 24, \text{wt}(\Delta_5) = 5$	native $(0, 0, \star, 5, 24)$; Open entry 3	realization package H_B across Open and CY quadrants	four native entries exact; Open entry conditional

10.2 The census-scoped archetypes

The atlas uses the labels

$$G, L, C, M, B.$$

Rows G, L, C, M refer to the finite Vol I standard-family census. Row B is the conditional $K_3 \times E$ comparison row. Their selected presentation depths

$$2, 3, 4, \infty, 5$$

belong to the selected shadow presentations. The Vol I scalar depth and the Vol III Borchers comparison depth are separate constructions. A family receives an archetype label after its residue projection and recurrence data have been constructed.

Remark 10.1 (Five measurements). The five coordinates

$$\kappa_{\text{cat}}, \kappa_{\text{ch}}^{\text{Hodge}}, \kappa_{\text{ch}}^{\text{Heis}}, \kappa_{\text{BKM}}, \kappa_{\text{fiber}}$$

have distinct sources and targets. A row in the 5×5 table is a tuple of comparisons among these measurements. The $K_3 \times E$ coordinate records a categorical value, two chiral values, a Borchers weight, and a topological Euler characteristic.

10.3 The Virasoro scalar lane

For $c(5c + 22) \neq 0$, the exact local seeds are

$$R_2 = \frac{c}{2}, \quad R_3 = 2, \quad R_4^{\text{Gram}} = \frac{10}{c(5c + 22)}.$$

On the domain $c(5c + 22) \neq 0$, the weighted unordered-pair recurrence gives

$$R_5^{\text{Ricc}} = -\frac{48}{c^2(5c + 22)}$$

and the sixth weighted-Riccati coefficient

$$R_6^{\text{Ricc}} = \frac{80(45c + 193)}{3c^3(5c + 22)^2}.$$

Imposing the formal quadratic relation

$$2R_2 C_6^{\text{rel}} + 2R_3 R_5 + R_4^2 = 0$$

with $R_2 = c/2$, $R_3 = 2$, $R_4 = 10/[c(5c + 22)]$, and $R_5 = -48/[c^2(5c + 22)]$, determines the order-six relation coefficient

$$C_6^{\text{rel}} = \frac{4(240c + 1031)}{c^3(5c + 22)^2}.$$

A singular-vector interpretation is supplied by an explicit level-six radical vector and its decoupling map to the chosen quotient or residue complex. A datum $H_{\text{res}}(\text{Vir}_c; X)$ consists of an ordered residue complex, an Arnold class, a scalar projection, and its normalization. This datum compares connected Ward correlators, Gram coefficients, the weighted recurrence, and the formal relation coefficient in a common scalar ambient.

For $45c + 218 \neq 0$, the discriminant of the weighted generating series has a simple zero and hence a square-root branch point. Its coefficients have the corresponding geometric growth with the power-law correction supplied by singularity analysis.

10.4 Family-indexed Hochschild support

For every chart $\mathcal{A}$, Theorem H begins with a complete complex $Q_{\mathcal{A}}$, a support model $K_{\mathcal{A},S}$, and a strong deformation retract. The resulting Hilbert polynomial, when the cohomology is finite-dimensional, is

$$P_{\mathcal{A}}(t) = \sum_{n \in S} \dim H^n(K_{\mathcal{A},S}) t^n.$$

Two bounded calculations provide exact benchmarks:

$$(\dim H_{\text{ch},b}^n(B_{\mathfrak{h}}))_{n \geq 0} = (2, 1, 0, \dots), \quad \text{Supp } H_{\text{ch},b}^{\bullet}(\text{Vir}_c) = \{0, 2, 3\},$$

with one-dimensional Virasoro groups in those three degrees [4, Theorems 7.4 and 7.2]. Each chart-level use of these vectors names its bounded-to-chart quasi-isomorphism. Affine, principal $\mathcal{W}$, lattice, free-field, and critical-level rows carry their respective data $H_H(\mathcal{A}; S_{\mathcal{A}})$.

10.5 Provenance rule for every row

A numerical row is assembled in the following order:

OPE/root/character calculation $\longrightarrow$ chain model $\longrightarrow$ comparison quasi-isomorphism $\longrightarrow$ global invariant.

The source calculation carries status *computed* or *proved elsewhere*. The comparison carries status *proved*, *conditional*, or *open*. The global row inherits the status of its comparison. This ordering makes the landscape a reproducible atlas of mathematical constructions.

References

- [1] V. I. Arnold, The cohomology ring of the colored braid group, *Mat. Zametki* **5** (1969), 227–231.
- [2] A. Beilinson and V. Drinfeld, *Chiral Algebras*, AMS Colloquium Publications, vol. 51, 2004.
- [3] A. Beilinson, V. Ginzburg, and W. Soergel, Koszul duality patterns in representation theory, *J. Amer. Math. Soc.* **9** (1996), no. 2, 473–527.
- [4] B. Bakalov, A. De Sole, and V. G. Kac, *Computation of cohomology of vertex algebras*, Japan. J. Math. **16** (2021), 81–154, arXiv:2002.03612.
- [5] R. E. Borcherds, Automorphic forms with singularities on Grassmannians, *Invent. Math.* **132** (1998), no. 3, 491–562.
- [6] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory*, vols. 1–2, Cambridge University Press, 2017–2021.
- [7] J. Francis and D. Gaitsgory, Chiral Koszul duality, *Selecta Math. (N.S.)* **18** (2012), no. 1, 27–87.
- [8] W. Fulton and R. MacPherson, A compactification of configuration spaces, *Ann. of Math. (2)* **139** (1994), no. 1, 183–225.
- [9] C. Faber and R. Pandharipande, Hodge integrals, partition matrices, and the λ_g conjecture, *Ann. of Math. (2)* **157** (2003), no. 1, 97–124.
- [10] E. Frenkel and D. Ben-Zvi, *Vertex Algebras and Algebraic Curves*, second edition, Mathematical Surveys and Monographs, vol. 88, American Mathematical Society, 2004.
- [11] H. Garland and J. Lepowsky, Lie algebra homology and the Macdonald–Kac formulas, *Invent. Math.* **34** (1976), 37–76.
- [12] E. Getzler and M. M. Kapranov, Modular operads, *Compositio Math.* **110** (1998), 65–126.
- [13] J. Gui, H. Li, and J. Zeng, Quadratic duality for chiral algebras, *Adv. Math.* **451** (2024), Paper no. 109791.
- [14] V. Gritsenko and V. Nikulin, The Igusa modular forms and “the simplest” Lorentzian Kac–Moody algebras, *Mat. Sb.* **187** (1996), no. 11, 27–66; arXiv:alg-geom/9603010.
- [15] J.-L. Loday and B. Vallette, *Algebraic Operads*, Grundlehren der mathematischen Wissenschaften, vol. 346, Springer, 2012.
- [16] S. C. Mok, Logarithmic Fulton–MacPherson configuration spaces, arXiv:2503.17563, 2025.
- [17] S. Mukai, On the moduli space of bundles on K_3 surfaces, I, in *Vector Bundles on Algebraic Varieties*, Tata Institute of Fundamental Research Studies in Mathematics, vol. 11, 1987, 341–413.
- [18] D. Mumford, Towards an enumerative geometry of the moduli space of curves, in *Arithmetic and Geometry*, vol. II, Birkhäuser, 1983, 271–328.
- [19] G. Faltings, Calculus on arithmetic surfaces, *Ann. of Math. (2)* **119** (1984), 387–424.
- [20] F. Malikov and V. Schechtman, Homotopy chiral algebras, arXiv:2408.16787, 2024.
- [21] S. Priddy, Koszul resolutions, *Trans. Amer. Math. Soc.* **152** (1970), 39–60.
- [22] L. Positselski, *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, Mem. Amer. Math. Soc. **212** (2011), no. 996.

A Comparison morphisms for the five theorems

This appendix records the mathematical object consumed by each theorem and the construction that remains family-dependent.

Theorem A

The published ambient statement is enhanced bar–cobar equivalence for the reduced associative operad in the pro-nilpotent Ran category. The factorization restriction is the closure package H_{fact} : preservation of factorization equivalences, collision base change, units, and completed inverse limits. The Verdier branch is the dualizability package H_{VD} . A self-duality point additionally carries a map $\sigma_{\mathcal{A}}: \mathcal{A} \xrightarrow{\sim} K_X(\mathcal{A})$.

Theorem B

The family calculation is the cone

$$\text{Cone}(q_{\mathcal{A}}: \mathcal{A}^i \longrightarrow B_X(\mathcal{A})).$$

The package H_{CL} builds this cone on finite conformal weight windows, contracts its positive-weight part compatibly with collision incidences and bar faces, and passes to the completed inverse limit. This construction proves the three intrinsic twisting conditions of Theorem B. The detecting PBW package $H_{\text{PBW}}^{\text{det}}$ adds the equivalent E_2 -collapse and diagonal-support condition.

Theorem C

A row of the 5×5 matrix is a tuple of five measurements. Every equality between two cells is represented by a comparison map. The Open–CY row B carries the realization package H_{B} ; its exact native entries are $(0, 0, \star, 5, 24)$, with $e_{\text{top}}(K_3) = 24$ and $\text{wt}(\Delta_5) = 5$. The comparison $\eta_{K_3, E}$ supplies the Open Heisenberg entry 3.

Theorem D

The hierarchy is

$$\text{Obs}_g^{\text{def}} \xrightarrow{H_D^{\text{r}}} \mathfrak{D}_g^K \xrightarrow{\text{ch}_g} \text{obs}_g^{\text{Hdg}}, \quad \text{Obs}_g^{\text{def}} \xrightarrow{H_D^{\text{graph}}} F_g.$$

The comparison H_D^K identifies $\mathfrak{D}_g^K$ with $\kappa\lambda_{-1}(\mathbb{E}_g)$. The Chern-root calculation then gives $(-1)^g \kappa\lambda_g$. The graph functional gives $\kappa\lambda_g^{\text{FP}} + \partial F_g^{\text{cross}}$.

Theorem H

The family datum $H_H(\mathcal{A}; S)$ is the strong deformation retract

$$K_{\mathcal{A}, S} \xrightarrow{\iota} Q_{\mathcal{A}} \xrightarrow{p} K_{\mathcal{A}, S}, \quad dh + hd = \text{id}_{Q_{\mathcal{A}}} - \iota p.$$

It includes the chart quasi-isomorphism, completion, and ordered-to-symmetric descent. Bounded vertex-cohomology calculations enter through a separate bounded-to-chart quasi-isomorphism.

B Exact computational layer

The Virasoro Ward recursion has been evaluated exactly through arity six. At central charge $c = 1$ and coordinates $(z_1, \dots, z_n) = (0, \dots, n-1)$, the connected values at arities five and six are

$$G_5^{\text{conn}} = \frac{775}{5184}, \quad G_6^{\text{conn}} = \frac{49705}{373248}.$$

Möbius inversion on set partitions and the free-boson cycle expansion give the same rational functions. Their passage to a scalar S_r is the residue map H_{res} .

On the domain $c(5c + 22) \neq 0$, the weighted unordered-pair recurrence gives

$$R_5^{\text{Ricc}} = -\frac{48}{c^2(5c + 22)}, \quad R_6^{\text{Ricc}} = \frac{80(45c + 193)}{3c^3(5c + 22)^2}.$$

The formal relation construction gives

$$C_6^{\text{rel}} = \frac{4(240c + 1031)}{c^3(5c + 22)^2}.$$

These are exact outputs of two named formal constructions.

For exceptional Yangians, exact computation presently covers root systems, Casimir eigenvalues, and tensor-square decompositions. A Yang–Baxter evaluation requires triple-tensor projectors and recoupling data. For Drinfeld–Sokolov reduction, exact computation covers type- $\mathcal{A}$ root arithmetic, nilpotent-orbit dimensions, and the $\mathfrak{sl}_2$ central-charge formulas. The induced chiral Hochschild map requires an explicit BRST chain map and homotopy.

C Open construction problems

- (O1) Construct the Heisenberg finite-window collision contraction in all strata and prove its compatibility with the inverse system.
- (O2) Construct $H_{\text{res}}(\text{Vir}_c; X)$. Its residue map defines $S_6(\text{Vir}_c; H_{\text{res}})$ and compares this geometric scalar with the weighted-Riccati and formal relation coefficients. A singular-vector comparison identifies its null-state specializations.
- (O3) Construct the family data $H_H(\mathcal{A}; \mathcal{S}_{\mathcal{A}})$ for affine and principal $\mathcal{W}$ charts, including their bounded-to-chart maps.
- (O4) Construct $\beta_g(\mathcal{A})$ in H_D^K from the scalar virtual bar object to $\kappa(\mathcal{A})\lambda_{-1}(\mathbb{E}_g)$, and construct the mixed-channel stable-graph functional.
- (O5) Construct the Open–CY realization package H_B for the $K_3 \times E$ row and compare its five coordinates in their native categories.
- (O6) Construct the triple-tensor recoupling data for exceptional Yangians and the BRST-induced chain maps for Drinfeld–Sokolov reduction.

Each problem names a source, a target, and the map whose construction advances the corresponding theorem.